

Problem Set 13

Integration by Parts I

October 2, 2026

Problem 2

(a)

$$\int x e^x dx = \left[\begin{array}{cc} e^x & x dx \\ e^x dx & \frac{1}{2} x^2 \end{array} \right] = \frac{e^x}{2} x^2 - \int \frac{e^x}{2} x^2 dx$$

The integral is harder to solve.

(b)

$$\begin{aligned} \int x^3 e^{x^2} dx &= \left[\begin{array}{cc} x^2 & x e^{x^2} dx \\ 2x dx & \frac{1}{2} e^{x^2} \end{array} \right] \frac{1}{2} x^2 e^{x^2} - \int x e^{x^2} dx \\ \int x e^{x^2} dx &= \frac{1}{2} e^{x^2} \\ \int x^3 e^{x^2} dx &= \boxed{\frac{1}{2} (x^2 - 1) e^{x^2} + C} \end{aligned}$$

(c)

$$\begin{aligned} \int u dv &= u(v + C) - \int v + C du = uv + uC - \int v du - uC = uv - \int v du \\ \int \ln(x+1) dx &= \left[\begin{array}{cc} \ln(x+1) & dx \\ \frac{1}{x+1} dx & x+1 \end{array} \right] (x+1) \ln(x+1) - \int 1 dx \\ &= \boxed{(x+1) \ln(x+1) - x + C} \end{aligned}$$

(d) The integrals on each side could hold different constants of integration, thereby differing the numerical value of each side of the equation by one.

Problem 3

(a) The two areas under the curve equal the area formed by the large rectangle subtracting the small rectangle.

(b)

$$\begin{aligned} u &= x = \sin y \\ v &= y = \arcsin x \\ \int_0^1 \arcsin x dx + \int_0^{\pi/2} \sin y dy &= \frac{\pi}{2} \\ \int_0^1 \arcsin x dx &= \frac{\pi}{2} - \int_0^{\pi/2} \sin y dy = \boxed{\frac{\pi}{2} - 1} \end{aligned}$$

(c)

$$\int_1^e \ln x \, dx = e - \int_0^1 e^x \, dx = \boxed{1}$$

Problem 4

$$\begin{aligned} f_{ave} &= \frac{\int_1^3 x^2 \ln x \, dx}{2} \\ \int_1^3 x^2 \ln x \, dx &= \left[\begin{array}{cc} \ln x & x^2 \, dx \\ \frac{1}{x} \, dx & \frac{1}{3} x^3 \end{array} \right] \frac{1}{3} x^3 \ln x \Big|_1^3 - \int_1^3 \frac{1}{3} x^2 \, dx \\ &= 9 \ln 3 + \frac{1}{9} - 3 = 9 \ln 3 - \frac{26}{9} \\ f_{ave} &= \frac{1}{2} \left(9 \ln 3 - \frac{26}{9} \right) = \boxed{\frac{9 \ln 3}{2} - \frac{13}{9}} \end{aligned}$$

Problem 5

(a)

$$\int x^2 \sin x \, dx = \boxed{-x^2 \cos x + 2x \sin x + 2 \cos x + C}$$

(b)

$$\begin{aligned} \int e^x \sin x \, dx &= e^x \sin x - e^x \cos x - \int e^x \sin x \, dx \\ \int e^x \sin x \, dx &= \boxed{\frac{e^x (\sin x - \cos x)}{2} + C} \end{aligned}$$

(c)

$$\begin{aligned} \int x^3 e^{2x} \, dx &= x^3 \cdot \frac{1}{2} e^{2x} - 3x^2 \cdot \frac{1}{4} e^{2x} + 6x \cdot \frac{1}{8} e^{2x} - 6 \cdot \frac{1}{16} e^{2x} \\ &= \boxed{\frac{e^{2x}}{8} (4x^3 - 6x^2 + 6x - 3) + C} \end{aligned}$$

(d) The table in (c) stops, as differentiating a polynomial will reach 0 eventually, while exponential and trig functions perpetuate. The method of (b) utilizes the fact that the original integral will appear on the other side after a finite number of integrations by parts.

Problem 6

(c)

$$\int f^{-1}(x) \, dx = \left[\begin{array}{cc} f^{-1}(x) & dx \\ (f^{-1})'(x) \, dx & x \end{array} \right] x f^{-1}(x) - \int x (f^{-1})'(x) \, dx$$

$$\int x(f^{-1})'(x) dx = \int f(y) dy = F(y) = F(f^{-1}(x))$$

$$\int f^{-1}(x) dx = \boxed{xf^{-1}(x) - F(f^{-1}(x)) + C}$$

(d)

$$f(x) = e^x \implies f^{-1}(x) = \ln x$$

$$\int \ln x dx = x \ln x - e^{\ln x} + C = \boxed{x \ln x - x + C}$$

Problem 7

(b)

$$\int \sin x \cos 4x dx = \left[\begin{array}{cc} \sin x & \cos 4x dx \\ \cos x dx & \frac{1}{4} \sin 4x \end{array} \right] \frac{1}{4} \sin x \sin 4x - \frac{1}{4} \int \cos x \sin 4x dx$$

$$= \left[\begin{array}{cc} \cos x & \sin 4x dx \\ -\sin x dx & -\frac{1}{4} \cos 4x \end{array} \right] \frac{1}{4} \sin x \sin 4x + \frac{1}{16} \cos x \cos 4x + \frac{1}{16} \int \sin x \cos 4x dx$$

$$\frac{15}{16} \int \sin x \cos 4x dx = \frac{1}{4} \sin x \sin 4x + \frac{1}{16} \cos x \cos 4x$$

$$\int \sin x \cos 4x dx = \boxed{\frac{4}{15} \sin x \sin 4x + \frac{1}{15} \cos x \cos 4x + C}$$

If the question is different, the u -sub may be applied.

(c)

$$\int \frac{xe^{2x}}{(2x+1)^2} dx = \left[\begin{array}{cc} xe^{2x} & (2x+1)^{-2} dx \\ e^{2x}(2x+1) dx & -\frac{1}{2(2x+1)} \end{array} \right] - \frac{xe^{2x}}{4x+2} + \int \frac{e^{2x}}{2} dx$$

$$= -\frac{xe^{2x}}{4x+2} + \frac{e^{2x}}{4} + C = \boxed{\frac{e^{2x}}{8x+4} + C}$$

Problem 8

(a)

$$\int \sec x dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$$

$$u = \sec x + \tan x$$

$$du = \sec^2 x + \sec x \tan x dx$$

$$\int \frac{du}{u} = \ln |u| = \boxed{\ln |\sec x + \tan x| + C}$$

(b)

$$\begin{aligned}
\int \sec^3 x \, dx &= \left[\begin{array}{cc} \sec x & \sec^2 x \, dx \\ \sec x \tan x \, dx & \tan x \end{array} \right] \sec x \tan x - \int \sec x \tan^2 x \, dx \\
&= \sec x \tan x - \int \sec^3 x - \sec x \, dx \\
2 \int \sec^3 x \, dx &= \sec x \tan x + \ln |\sec x + \tan x| \\
\int \sec^3 x \, dx &= \boxed{\frac{1}{2} (\sec x \tan x + \ln |\sec x + \tan x|) + C}
\end{aligned}$$

Problem 9

(a)

$$\begin{aligned}
J_n &= \int_0^1 x^n e^{-x} \, dx = \left[\begin{array}{cc} x^n & e^{-x} \, dx \\ nx^{n-1} \, dx & -e^{-x} \end{array} \right] - x^n e^{-x} \Big|_0^1 + n \int_0^1 x^{n-1} e^{-x} \, dx = \boxed{n J_{n-1} - \frac{1}{e}} \\
J_0 &= \int_0^1 x^0 e^{-x} \, dx = \int_0^1 e^{-x} \, dx = -e^{-x} \Big|_0^1 = \boxed{1 - \frac{1}{e}}
\end{aligned}$$

(b) *Proof.*

$$J_n = n! - \frac{n!}{e} \sum_{k=0}^n \frac{1}{k!}, \quad n \geq 0$$

Base case ($n = 0$):

$$J_0 = 0! - \frac{0!}{e} \sum_{k=0}^0 \frac{1}{k!} = 1 - \frac{1}{e} \cdot \frac{1}{1} = 1 - \frac{1}{e}$$

Inductive step:

Assume the statement holds for $n = m$, $m \geq 0$:

$$J_m = m! - \frac{m!}{e} \sum_{k=0}^m \frac{1}{k!}$$

We now show that the statement holds for $n = m + 1$:

$$\begin{aligned}
J_{m+1} &= (m+1)J_m - \frac{1}{e} \\
&= (m+1) \left(m! - \frac{m!}{e} \sum_{k=0}^m \frac{1}{k!} \right) - \frac{1}{e} \\
&= (m+1)m! - \frac{(m+1)m!}{e} \sum_{k=0}^m \frac{1}{k!} - \frac{1}{e} \\
&= (m+1)m! - \frac{(m+1)m!}{e} \left(\sum_{k=0}^m \frac{1}{k!} + \frac{1}{(m+1)!} \right)
\end{aligned}$$

$$= (m+1)! - \frac{(m+1)!}{e} \sum_{k=0}^{m+1} \frac{1}{k!}$$

Hence, the statement holds for $n = m + 1$. Therefore, by the Principle of Mathematical Induction:

$$J_n = n! - \frac{n!}{e} \sum_{k=0}^n \frac{1}{k!}, \quad n \geq 0$$

(c)

$$\begin{aligned} J_n &= \int_0^1 x^n e^{-x} dx \\ x^n, e^{-x} &> 0 \implies J_n > 0 \\ J_n &= \int_0^1 x^n e^{-x} dx < \int_0^1 x^n dx = \left. \frac{x^{n+1}}{n+1} \right|_0^1 = \frac{1}{n+1} \\ \boxed{0 < J_n < \frac{1}{n+1}} \end{aligned}$$

(d)

$$\begin{aligned} \text{let } e &= \frac{p}{q}, n \geq \max(q, 2) \\ J_n &= n! - \frac{n!}{e} \sum_{k=0}^n \frac{1}{k!} \\ e J_n &= e \cdot n! - n! \sum_{k=0}^n \frac{1}{k!} \\ e J_n &= \left[\frac{p}{q} \cdot n! - n! \sum_{k=0}^n \frac{1}{k!} \right] \\ 0 &< J_n < \frac{1}{n+1} \\ 0 &< e J_n < \frac{e}{n+1} \\ n \geq 2 &\implies n+1 \geq 3 \implies \frac{e}{n+1} < 1 \\ \boxed{0 < e J_n < 1} \end{aligned}$$

As the value of $e J_n$ is within this range, it cannot be an integer, thus contradicting the assumption that e is rational.

(e)

$$J_n = n! - \frac{n!}{e} \sum_{k=0}^n \frac{1}{k!}$$

$$\sum_{k=0}^n \frac{1}{k!} = e - \frac{e J_n}{n!}$$

$$0 < e J_n < 1 \implies \lim_{n \rightarrow \infty} \frac{e J_n}{n!} = 0$$

$$\boxed{\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{1}{k!} = e}$$